



MODULUS OF COMPLEX NUMBERS



EXAMPLES



Example 2.9

If $z_1 = 3 + 4i$, $z_2 = 5 - 12i$, and $z_3 = 6 + 8i$,
find $|z_1|$, $|z_2|$, $|z_3|$, $|z_1 + z_2|$, $|z_2 - z_3|$, and $|z_1 + z_3|$.

Solution

We use $|a + bi| = \sqrt{a^2 + b^2}$ to find the modulus.

1 Find $|z_1|$

$$|z_1| = |3 + 4i| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

2 Find $|z_2|$

$$|z_2| = |5 - 12i| = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$$

3 Find $|z_3|$

$$|z_3| = |6 + 8i| = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$$

4 Find $|z_1 + z_2|$

$$z_1 + z_2 = (3 + 4i) + (5 - 12i) = 8 - 8i$$

$$\begin{aligned} |z_1 + z_2| &= |8 - 8i| = \sqrt{8^2 + (-8)^2} \\ &= \sqrt{64 + 64} = \sqrt{128} = 8\sqrt{2} \end{aligned}$$

5 Find $|z_2 - z_3|$

$$z_2 - z_3 = (5 - 12i) - (6 + 8i) = -1 - 20i$$

$$\begin{aligned} |z_2 - z_3| &= |-1 - 20i| = \sqrt{(-1)^2 + (-20)^2} \\ &= \sqrt{1 + 400} = \sqrt{401} \end{aligned}$$

6 Find $|z_1 + z_3|$

$$z_1 + z_3 = (3 + 4i) + (6 + 8i) = 9 + 12i$$

$$\begin{aligned} |z_1 + z_3| &= |9 + 12i| = \sqrt{9^2 + 12^2} \\ &= \sqrt{81 + 144} = \sqrt{225} = 15 \end{aligned}$$

Final Answers

$$|z_1| = 5 \quad |z_2| = 13 \quad |z_3| = 10 \quad |z_1 + z_2| = 8\sqrt{2} \quad |z_2 - z_3| = \sqrt{401} \quad |z_1 + z_3| = 15$$

Example 2.10 (i)

Problem

$$\left| \frac{2+i}{-1+2i} \right|$$



Solution

Concept Used: $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}, \quad z_2 \neq 0$

1

Find the modulus of the numerator

$$|2+i| = \sqrt{2^2 + 1^2} = \sqrt{4+1} = \sqrt{5}$$

2

Find the modulus of the denominator

$$|-1+2i| = \sqrt{(-1)^2 + (2)^2} = \sqrt{1+4} = \sqrt{5}$$

3

Divide the moduli

$$\left| \frac{2+i}{-1+2i} \right| = \frac{|2+i|}{|-1+2i|} = \frac{\sqrt{5}}{\sqrt{5}} = 1$$

Answer: 1



Example 2.10 (ii)

Problem

$$|(1+i)(2+3i)(4i-3)|$$



Solution

Concept Used: $|z_1 z_2 z_3| = |z_1| |z_2| |z_3|$

1

Find $|1+i|$

$$|1+i| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

2

Find $|2+3i|$

$$|2+3i| = \sqrt{2^2 + 3^2} = \sqrt{13}$$

3

Find $|4i-3|$

$$|4i-3| = |-3+4i| = \sqrt{(-3)^2 + 4^2} = \sqrt{9+16} = 5$$

4

Multiply the moduli

$$|(1+i)(2+3i)(4i-3)| = (\sqrt{2})(\sqrt{13})(5) = 5\sqrt{26}$$

Answer: $5\sqrt{26}$



Example 2.10 (iii)

Problem

$$\left| \frac{(2+i)^3}{(1+i)^2} \right|$$



Solution

Concept Used: $\left| \frac{z_1^m}{z_2^n} \right| = \frac{|z_1|^m}{|z_2|^n}, \quad z_2 \neq 0$

1

Find $|2+i|$

$$|2+i| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

2

Find $|1+i|$

$$|1+i| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

3

Apply the formula

$$\left| \frac{(2+i)^3}{(1+i)^2} \right| = \frac{|2+i|^3}{|1+i|^2} = \frac{(\sqrt{5})^3}{(\sqrt{2})^2}$$

4

Simplify

$$\frac{(\sqrt{5})^3}{(\sqrt{2})^2} = \frac{5\sqrt{5}}{2}$$

Answer:

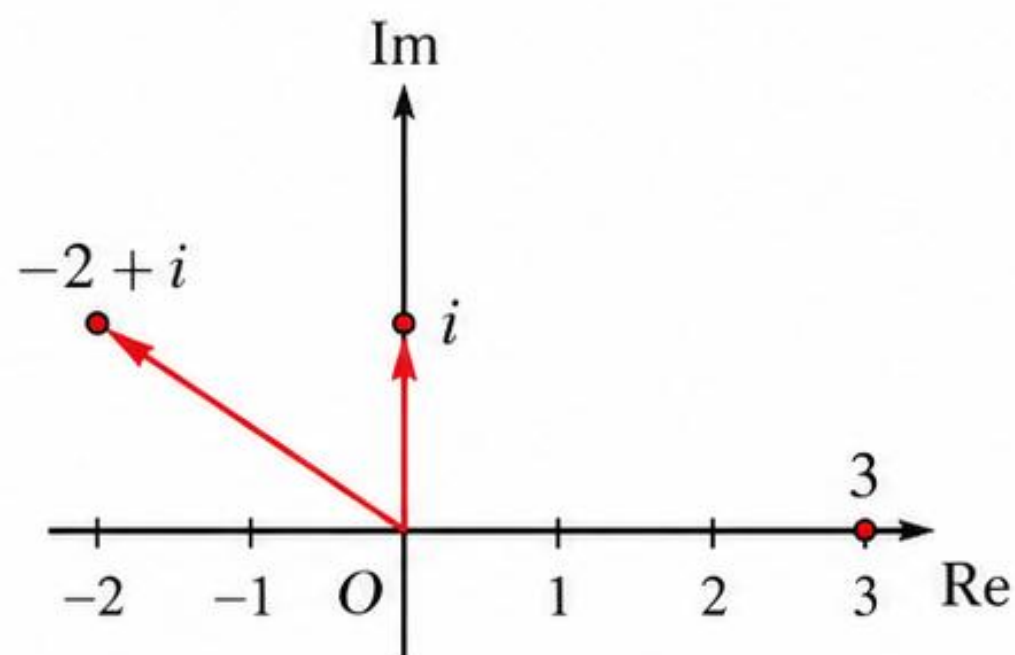
$$\frac{5\sqrt{5}}{2}$$



Example 2.11

Problem

Which one of the points i , $-2 + i$, and 3 is farthest from the origin?



Solution

The distance of a complex number z from the origin is $|z|$.
Let us find the distance of each point from the origin.

1 For i : $|i| = |0 + 1i| = \sqrt{0^2 + 1^2} = \sqrt{1} = \boxed{1}$

2 For $-2 + i$: $|-2 + i| = |-2 + 1i| = \sqrt{(-2)^2 + 1^2}$
 $= \sqrt{4 + 1} = \boxed{\sqrt{5}}$

3 For 3 : $|3| = |3 + 0i| = \sqrt{3^2 + 0^2} = \sqrt{9} = \boxed{3}$

Comparison: $1 < \sqrt{5} < 3 \Rightarrow$ The farthest point from the origin is **3**.

Example 2.12

Problem

If z_1 , z_2 , and z_3 are complex numbers such that

$$|z_1| = |z_2| = |z_3| = |z_1 + z_2 + z_3| = 1,$$

find the value of

$$\left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right|.$$

Solution

Since $|z_1| = |z_2| = |z_3| = 1$,
we know that for any complex number z with $|z| = 1$,

$$\bar{z} = \frac{1}{z}.$$

→ Hence, $\frac{1}{z_1} = \bar{z}_1$, $\frac{1}{z_2} = \bar{z}_2$, $\frac{1}{z_3} = \bar{z}_3$.

→ Substituting in the given expression,

$$\left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right| = |\bar{z}_1 + \bar{z}_2 + \bar{z}_3|$$

→ But $\bar{z}_1 + \bar{z}_2 + \bar{z}_3 = \overline{z_1 + z_2 + z_3}$

→ Therefore,
$$\left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right| = |\overline{z_1 + z_2 + z_3}| = |z_1 + z_2 + z_3| = 1.$$

Answer: 1

Example 2.13

Problem

If $|z| = 2$, show that
 $3 \leq |z + 3 + 4i| \leq 7$

Concept Used

Triangle Inequality:

$$|w_1 + w_2| \leq |w_1| + |w_2|$$

$$|w_1 + w_2| \geq \left| |w_1| - |w_2| \right|$$

Let $w_1 = z$ and $w_2 = 3 + 4i$.

Then $|w_1| = |z| = 2$ and $|w_2| = |3 + 4i| = 5$.

Solution

Upper Bound:

Using $|w_1 + w_2| \leq |w_1| + |w_2|$,

$$|z + 3 + 4i| \leq |z| + |3 + 4i| = 2 + 5 = \boxed{7}$$

Lower Bound:

Using $|w_1 + w_2| \geq \left| |w_1| - |w_2| \right|$,

$$|z + 3 + 4i| \geq \left| |z| - |3 + 4i| \right| = |2 - 5| = \boxed{3}$$

Conclusion:

Combining both bounds,

$$3 \leq |z + 3 + 4i| \leq 7$$

Example 2.14

Show that the points 1 , $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$, and $-\frac{1}{2} - i\frac{\sqrt{3}}{2}$, are the vertices of an equilateral triangle.

► Let

$$z_1 = 1, \quad z_2 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}, \quad z_3 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$$

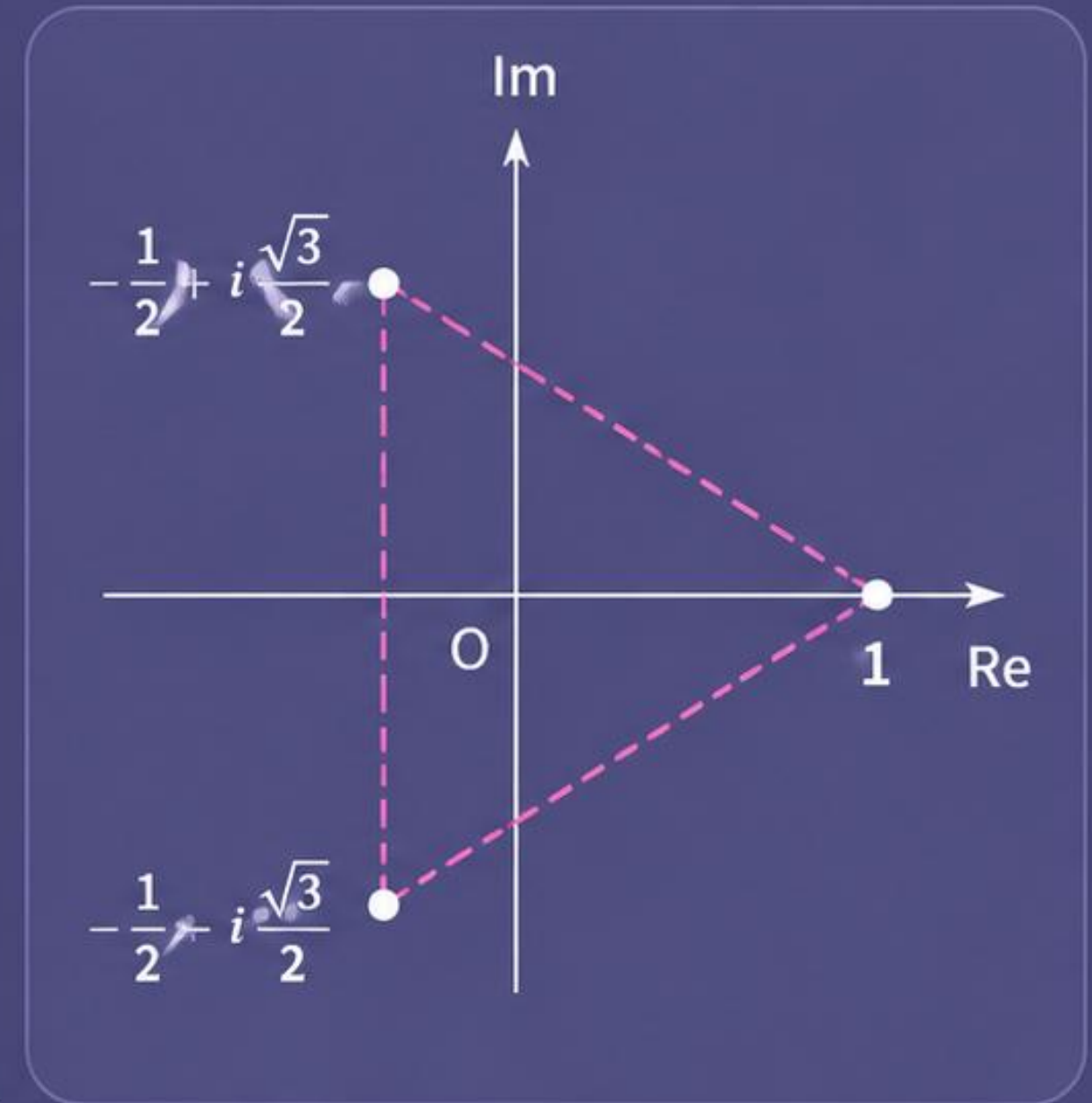
► Length of the sides

$$\textcircled{1} \quad |z_1 - z_2| = \left| 1 - \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) \right| = \left| \frac{3}{2} - i\frac{\sqrt{3}}{2} \right| = \sqrt{\left(\frac{3}{2} \right)^2 + \left(\frac{\sqrt{3}}{2} \right)^2} = \sqrt{3}$$

$$\textcircled{2} \quad |z_2 - z_3| = \left| \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) - \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right) \right| = |i\sqrt{3}| = \sqrt{3}$$

$$\textcircled{3} \quad |z_3 - z_1| = \left| \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right) - 1 \right| = \left| -\frac{3}{2} - i\frac{\sqrt{3}}{2} \right| = \sqrt{\left(\frac{3}{2} \right)^2 + \left(\frac{\sqrt{3}}{2} \right)^2} = \sqrt{3}$$

★ Since $|z_1 - z_2| = |z_2 - z_3| = |z_3 - z_1| = \sqrt{3}$,
all three sides are equal. Therefore, the given points form an equilateral triangle.



Example 2.15

Let z_1, z_2, z_3 be complex numbers such that $|z_1| = |z_2| = |z_3| = r > 0$ and $z_1 + z_2 + z_3 \neq 0$.

Prove that $\left| \frac{z_1 z_2 + z_2 z_3 + z_3 z_1}{z_1 + z_2 + z_3} \right| = r$.

Solution:

1 Use modulus property

$$\begin{aligned} \text{Since } |z_k| = r, \text{ we have } |z_k|^2 = r^2 = z_k \bar{z}_k \quad (k = 1, 2, 3). \\ \Rightarrow z_1 = \frac{r^2}{\bar{z}_1}, \quad z_2 = \frac{r^2}{\bar{z}_2}, \quad z_3 = \frac{r^2}{\bar{z}_3}. \end{aligned}$$

2 Write $z_1 + z_2 + z_3$

$$\begin{aligned} z_1 + z_2 + z_3 &= \frac{r^2}{\bar{z}_1} + \frac{r^2}{\bar{z}_2} + \frac{r^2}{\bar{z}_3} \\ &= r^2 \left(\frac{\bar{z}_2 \bar{z}_3 + \bar{z}_3 \bar{z}_1 + \bar{z}_1 \bar{z}_2}{\bar{z}_1 \bar{z}_2 \bar{z}_3} \right) \end{aligned}$$

3 Take modulus on both sides

$$|z_1 + z_2 + z_3| = \left| r^2 \frac{\bar{z}_2 \bar{z}_3 + \bar{z}_3 \bar{z}_1 + \bar{z}_1 \bar{z}_2}{\bar{z}_1 \bar{z}_2 \bar{z}_3} \right| \quad (\because z_1 + z_2 + z_3 \neq 0)$$

4 Use modulus properties

$$\begin{aligned} |z_1 + z_2 + z_3| &= |r^2| \left| \frac{\bar{z}_2 \bar{z}_3 + \bar{z}_3 \bar{z}_1 + \bar{z}_1 \bar{z}_2}{\bar{z}_1 \bar{z}_2 \bar{z}_3} \right| \quad (\because \left| \frac{a}{b} \right| = \frac{|a|}{|b|}) \\ &= r^2 \left| \frac{\bar{z}_2 \bar{z}_3 + \bar{z}_3 \bar{z}_1 + \bar{z}_1 \bar{z}_2}{|\bar{z}_1| |\bar{z}_2| |\bar{z}_3|} \right| \quad (\because |\bar{z}| = |z|) \\ &= r^2 \left| \frac{z_2 z_3 + z_3 z_1 + z_1 z_2}{|z_1| |z_2| |z_3|} \right| \quad (\because |\bar{z}| = |z|) \end{aligned}$$

5 Use $|z_1| = |z_2| = |z_3| = r$

$$\begin{aligned} \Rightarrow |z_1 + z_2 + z_3| &= r^2 \frac{|z_2 z_3 + z_3 z_1 + z_1 z_2|}{r r r} \\ &= \frac{1}{r} |z_2 z_3 + z_3 z_1 + z_1 z_2| \end{aligned}$$

6 Remove conjugate inside modulus

$$\begin{aligned} \text{Since } |w| &= |\bar{w}|, \\ |z_1 + z_2 + z_3| &= \frac{1}{r} |z_2 z_3 + z_3 z_1 + z_1 z_2| \end{aligned}$$



Therefore,

$$\left| \frac{z_1 z_2 + z_2 z_3 + z_3 z_1}{z_1 + z_2 + z_3} \right| = r.$$

2.16 Show that the equation $z^2 = \bar{z}$ has four solutions.

Solution:

1 Take modulus on both sides

We have,

$$z^2 = \bar{z}$$

Taking modulus,

$$|z^2| = |\bar{z}|$$

$$\Rightarrow |z|^2 = |z|$$

$$\Rightarrow |z|(|z| - 1) = 0$$

$$\Rightarrow |z| = 0, \text{ or } |z| = 1.$$

2 Case 1: $|z| = 0$

If $|z| = 0$, then $z = 0$.

So, $z = 0$ is one solution.

3 Case 2: $|z| = 1$

If $|z| = 1$, then $z\bar{z} = 1 \Rightarrow \bar{z} = \frac{1}{z}$ ($z \neq 0$).

Substitute in $z^2 = \bar{z}$,

$$z^2 = \frac{1}{z} \Rightarrow z^3 = 1.$$

4 Solve $z^3 = 1$

The cube roots of unity are the solutions of $z^3 = 1$.
They are:

$$z_1 = 1, \quad z_2 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad z_3 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

5 Conclusion



There are three solutions from $z^3 = 1$ and one solution $z = 0$.

Hence, the equation $z^2 = \bar{z}$ has four solutions.

2.17**Find the square root of $6 - 8i$.****1 Write in the form $a + bi$** Let $z = 6 - 8i$.Here, $a = 6$ and $b = -8$.**2 Find $r = |z|$**

$$\begin{aligned}
 r = |z| &= \sqrt{a^2 + b^2} \\
 &= \sqrt{6^2 + (-8)^2} \\
 &= \sqrt{36 + 64} \\
 &= \sqrt{100} = 10
 \end{aligned}$$

Formula usedFor $z = a + bi$,

$$\sqrt{z} = \pm \left(\sqrt{\frac{r+a}{2}} + i \operatorname{sgn}(b) \sqrt{\frac{r-a}{2}} \right)$$

where $r = \sqrt{a^2 + b^2}$ and $\operatorname{sgn}(b) = \begin{cases} 1, & b > 0 \\ -1, & b < 0. \end{cases}$

3 Apply the formulaHere, $a = 6$, $b = -8 (< 0)$, $r = 10$. So,

$$\begin{aligned}
 \sqrt{6 - 8i} &= \pm \left(\sqrt{\frac{10+6}{2}} + i(-1) \sqrt{\frac{10-6}{2}} \right) \\
 &= \pm \left(\sqrt{\frac{16}{2}} - i \sqrt{\frac{4}{2}} \right) \\
 &= \pm (\sqrt{8} - i\sqrt{2})
 \end{aligned}$$

4 Simplify

$$\begin{aligned}
 \text{Since } \sqrt{8} &= \sqrt{4 \times 2} = 2\sqrt{2}, \\
 \sqrt{6 - 8i} &= \pm (2\sqrt{2} - i\sqrt{2}) \\
 &= \pm \sqrt{2} (2 - i) \\
 &= \pm (2\sqrt{2} - i\sqrt{2})
 \end{aligned}$$

Therefore,

$$\sqrt{6 - 8i} = \text{pm}(2\sqrt{2} - i\sqrt{2})$$



Thank You